

Joe F. Smith

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SUMMARY

Undergraduate researcher at the intersection of physics and artificial intelligence, applying deep learning to problems in black hole population models, mechanistic interpretability, and AI alignment. Seven papers, including sole authorship in *The Astrophysical Journal* and Neurips 2026 (x3).

EDUCATION

California Institute of Technology
B.S. Physics and Computer Science (GPA: 3.9)
Relevant Coursework: Relativistic Astrophysics, Quantum Physics (Mechanics, Quantum Computation), ML & Data Mining, Linear Algebra, Differential Equations, Probability & Statistics
Activities: Caltech AI Alignment (CAIA) ML Researcher, Quant Club, RISE Tutor, ARC Representative
Programs: Discover Citadel | Citadel Securities (Apr 2026); DE Shaw Connect (Aug 2026)

Pasadena, CA
Graduation: 06/28

RESEARCH EXPERIENCE

Kareem El-Badry Research Group @ Caltech
Astrophysics Undergraduate Researcher
• Programmed a high-performance computing (HPC) Python pipeline to test new physics describing mass transfer in ultracompact X-ray binaries.

Institute of Astronomy, University of Cambridge
Astrophysics and Machine Learning Research Intern
• Lead machine learning engineer under Prof. Vasily Belokurov studying the archaeology of the Magellanic Clouds; led architecture design, model implementation, figures, and science quality checks.
• Leveraged unsupervised and supervised methods (ensemble learning, variational autoencoders, GP classification) to differentiate galactic spectra, with data-driven visualizations comparing results against prior literature.
• Derived physics in MATLAB and developed Markov Chain Monte Carlo (MCMC) simulations, revealing statistically significant populations.

Pasadena, CA
Jan 2026 – Present
Cambridge, UK
May 2024 – Aug 2024

PROJECTS

Chain-of-Thought Invariance and Monitorability
• Built extensive pipelines for policy, rewriting, verifying, and monitoring agents for identifying sensitivity to linguistic frameworks to find when chain-of-thought monitoring does not notice harmful model behavior, emphasizing the effects of rewording and repositioning reasoning traces. Accepted to the workshops on Linguistic Principles for Foundation Models (LP4FM), Trustworthy AI for Good (AI4GOOD), and Trust-AI-Eval (TAE) @ Neurips 2026. Submitting to ICML 2027.
• Queried models via OpenRouter and Anthropic APIs accessing different LLM families for different roles and utilized HuggingFace to pull MMLU-Redux 2.0 and a public sandbagging benchmark for testing LLM monitor evasion.

Astrometric Binary Classification via Artificial Neural Networks
• Created the first automated astrometric binary classifier, published as sole author in *The Astrophysical Journal* with 3.0K+ article and 1.0K+ software downloads.
• Reduced manual classification time by >70% with 99.3% accuracy and 0.999 AUC for 1.5mm+ binaries on a low-cost CPU, enabling model deployment in seconds.
• Verified the architecture against benchmark models through statistical comparisons (e.g. DeLong test) and ablation tests with physical reasoning.

June 2026 — Present
Dec 2023 – Apr 2024

SELECTED PUBLICATIONS

Smith, J. “Output Equivalence Does Not Imply Causal Invariance in Chain-of-Thought Monitors.” LP4FM @ Neurips 2026, Sydney, AUS.
Smith, J. “Chain-of-Thought Restructuring Does Not Reduce Sandbagging Monitorability.” AI4GOOD @ Neurips 2026, Paris, FRA.
Smith, J. “Same Reasoning, Different Words: Testing Linguistic Invariance in Chain-of-Thought Monitoring.” TAE @ Neurips 2026, Sydney, AUS.
Smith, J. “Astrometric Binary Classification via Artificial Neural Networks.” *The Astrophysical Journal*, 974, 96 (2024). arXiv:2409.09563
Smith, J., Simpson, B., Boyce, P. “Speckle Observation of WDS 15493+6032 HU 912.” *Journal of Double Star Observations*, 19, 4, 409 (2023).

AWARDS

• Caltech Summer Undergraduate Research Fellowship (2026)
• ISEF Physics & Astronomy 3rd Grand Award (2025)

• Davidson Fellow (2025)
• AAVSO Exoplanet Observer Award (2023)

• Regeneron STS Scholar (2025)

SKILLS

Languages: Python, C, Java, R, MATLAB, Fortran, ADQL, SQL, SLURM
Libraries: PyTorch, TensorFlow, scikit-learn, emcee, lmfit, Pandas, NumPy, SciPy, Astropy, Matplotlib, seaborn
Reviewer: LP4FM, AutoMLR, MLP4S, AI4GOOD @ Neurips 2026 (7 papers)
Miscellaneous: Ham radio amateur extra (W6SEA), private glider pilot, rowing, chess, electric guitar

Developer Tools: Git, Jupyter, Google Colab, MESA, HuggingFace